

Fanuc 0i-TF Operator Manual — Alarms & Diagnostics

Site excerpt for installed base on DMU-14, DMU-22 (5-axis VMC) and NLX-08, NLX-09 (turning centers). For full OEM manual, see PDM record FANUC-0iTF-OEM-2023.

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§7.1 Scope

This excerpt covers the alarm-handling section of the Fanuc 0i-TF Operator Manual as deployed across Contoso Precision Parts production cells. The 0i-TF controller is installed on the DMG Mori DMU 50 5-axis machining centers (DMU-14 , DMU-22) and on the Mori Seiki NLX 2500 turning centers (NLX-08 , NLX-09). All alarm numbers in this excerpt correspond to the 0i-TF CNC system; servo and spindle alarms have their own numbering ranges and are documented in §7.4 and §7.5 of the full OEM manual.

§7.2 General alarm-clearance procedure

1. Note the alarm number and channel displayed on the operator screen.
2. Press **RESET**. If the alarm clears, log it in the shift handover and continue.
3. If **RESET** does not clear the alarm, look up the alarm in §7.3 (system alarms) or §7.4 (servo) and follow the documented corrective sequence.
4. For any alarm requiring access inside the work envelope or behind a guard, follow the machine-specific LOTO procedure before opening the enclosure.

§7.3 System alarms (PS / SR series)

Code	Display message	Probable causes	Corrective action
300	APC ALARM: nth-AXIS NEED ZRN (<i>Absolute Position Coder — reference return required</i>)	(a) APC backup battery low or replaced. (b) Encoder cable disconnected while controller was powered off. (c) Servo amplifier or motor replaced without subsequent reference return. (d) Power loss to encoder while machine moved. NOTE: This alarm is a position-feedback fault, not a tool-condition fault. Tool wear on the DMU-14/DMU-22 is tracked separately via the Inconel 718 process spec (PLM-PS-INC718-0042).	1. Confirm operator is clear of work envelope. 2. Switch mode selector to JOG or REF. 3. For each affected axis, perform manual reference return per machine setup guide. 4. Press RESET. Alarm 300 should clear. 5. If alarm recurs after re-homing: check APC battery voltage (≥ 3.0 V required) and replace if low. See service bulletin PDM-SB-DMU14-2024-03 for the DMU-14-specific replacement procedure. 6. If alarm persists after battery replacement, escalate to maintenance: encoder or cable replacement may be required.
401	SERVO VRDY OFF	Servo amplifier not in ready state. Common causes: E-stop pressed, 200 V phase loss, amplifier fault LED illuminated.	Release E-stop, confirm three-phase supply, check amplifier LED status. If LED indicates internal fault, replace amplifier per maintenance work order.
506	OVER TRAVEL: +X	Programmed move exceeded positive X soft limit. Work offset may be wrong, or part program may reference an out-of-range absolute coordinate.	Jog axis back inside the soft-limit range, verify work coordinate offsets, single-block through the offending move to confirm correct.

Cross-reference. Alarm 300 on DMU-14 specifically: see site service bulletin PDM-SB-DMU14-2024-03 . Tool-wear-related shutdowns on Inconel 718 are governed by process spec PLM-PS-INC718-0042 and surface as program messages rather than alarm 300.

§7.4 Servo alarms (SV series, summary)

Servo alarms 400–499 indicate amplifier, motor, or feedback faults. The most common on the 0i-TF installed base at Contoso are summarised below; for the full list see the OEM manual.

- **SV0410 / SV0411** — excessive position error during stop / motion. Check servo gain settings and mechanical load.
- **SV0414** — digital servo system alarm. Power-cycle the controller; if persistent, replace the servo amplifier.

\$7.5 When to call Process Engineering

Call Process Engineering immediately if: (a) the same alarm recurs more than twice in a single shift, (b) an alarm appears that is not documented in §7.3, §7.4 or §7.5, or (c) clearing the alarm would require opening a guarded enclosure beyond a quick door-interlock reset. For machine-specific escalation contacts, see the shift-handover board in each cell.

Document control / sign-off

Role	Name	Signature	Date
Document Owner	M. Tanaka, Process Engineering — Wichita	(on file)	2024-06-01
Approver	M. Tanaka, Process Engineering — Wichita	(on file)	2024-06-01